Low-cost Passive Acoustic Sensing of Bridge Cracks

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Abstract: We propose a passive acoustic method that locates hairline cracks in steel bridges using inexpensive microphones. The method needs no traffic closure and no trained inspector on site. In field trials on two bridges it matched the accuracy of ultrasonic scanning while running continuously for six months.

Keywords: acoustic sensing; structural health monitoring; steel bridges; crack detection

1. Introduction

The main challenge in bridge inspection is that ultrasonic scanning is slow and usually requires lane closures. Existing methods also give only a snapshot in time, so cracks that grow between inspections go unnoticed. Unlike existing methods, our approach runs continuously and unattended, and it costs a fraction of scanning.

2. Method

We attach low-cost microphones to girders and record the sound of passing traffic. A signal processing pipeline separates traffic noise from the faint acoustic signature of a crack opening and closing under load. We propose a new calibration procedure that learns each bridge’s baseline in two days. The pipeline runs on a small embedded board and sends a daily summary over a cellular link.

3. Results

In trials on two steel bridges we located 41 of 44 known cracks to within 30 centimeters. False alarms were rare. Early results were presented at the Midwest Sensing Workshop on June 12, 2026 and posted on a project website.

4. Limitations

A limitation is that the method does not work on concrete decks without further calibration. Performance also drops in heavy rain, and we have not tested it on bridges with complex cable structures.

Acknowledgments

This work was supported by the National Science Foundation under grant number 2012345 and by a seed grant from the university. We thank the state transportation department for access to the test bridges.

References

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All names are fictional.